

GMUNDEN, Austria

WMO: 111540

Lat: 47.915N

Lon: 13.807E

Elev: 428

StdP: 96.29

Time Zone: 1.00 (EUC)

Period: 94-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.6	-8.4	-12.6	1.3	-10.1	-10.2	1.6	-7.4	7.4	9.0	6.5	7.3	1.2	20	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	30.6	21.1	28.6	20.2	26.8	19.3	21.7	29.0	20.7	27.4	19.9	25.6	1.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.3	14.8	25.1	18.5	14.1	23.8	17.7	13.4	22.8	65.5	29.3	61.8	27.6	58.8	25.6	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.4	4.6	4.0	DB	-12.2	33.8	3.3	1.8	-14.6	35.1	-16.6	36.2	-18.4	37.2	-20.9	38.5	(4)
(5)			WB	-12.5	23.4	3.2	1.4	-14.8	24.4	-16.7	25.2	-18.5	26.0	-20.8	27.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-0.2	1.7	5.4	10.1	14.5	17.7	19.9	19.6	14.9	10.1	5.0	1.3	(6)	
	DBStd	8.06	4.71	4.61	4.09	4.15	3.73	3.91	3.38	3.41	3.31	3.87	3.98	4.44	(7)	
	HDD10.0	1231	316	233	148	50	6	1	0	0	2	48	156	271	(8)	
	HDD18.3	3218	573	466	399	249	128	59	22	25	111	257	400	529	(9)	
	CDD10.0	1247	1	1	7	52	146	231	307	298	149	50	6	1	(10)	
	CDD18.3	194	0	0	0	1	10	39	71	64	8	1	0	0	(11)	
	CDH23.3	1451	0	0	0	11	99	317	523	458	43	1	0	0	(12)	
	CDH26.7	422	0	0	0	1	15	86	173	141	6	0	0	0	(13)	
(14) Wind	WSAvg	1.6	2.0	1.9	1.8	1.6	1.5	1.5	1.6	1.4	1.3	1.3	1.6	1.7	(14)	
(15) Precipitation	PrecAvg	1448	96	80	113	89	135	164	173	159	136	99	101	102	(15)	
	PrecMax	1739	222	181	207	180	271	332	354	342	264	174	247	203	(16)	
	PrecMin	1141	7	26	45	21	31	85	60	61	52	31	10	34	(17)	
	PrecStd	143	52	41	53	38	54	52	65	55	56	41	49	41	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.0	18.9	24.7	28.6	31.6	33.3	33.2	27.1	21.8	18.0	12.3	(19)	
		MCWB	8.4	8.2	11.3	15.4	19.3	21.7	22.7	21.3	19.4	15.8	11.4	7.7	(20)	
	2%	DB	10.2	11.3	16.1	22.0	26.2	29.2	30.6	30.2	24.3	19.0	14.3	10.5	(21)	
		MCWB	7.1	7.3	10.0	14.1	18.1	20.7	21.4	20.6	17.9	14.4	10.2	7.4	(22)	
	5%	DB	8.2	9.4	13.6	19.7	23.9	27.2	28.5	27.9	22.1	17.1	12.1	8.7	(23)	
		MCWB	5.8	6.2	8.8	12.8	16.6	19.4	20.3	20.0	17.0	13.5	9.8	6.4	(24)	
	10%	DB	6.3	7.6	11.4	17.4	21.6	24.9	26.3	25.9	20.1	15.4	10.4	7.1	(25)	
		MCWB	4.4	5.1	7.6	11.5	15.2	18.1	19.4	19.1	15.9	12.7	8.4	5.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.0	9.2	11.9	16.2	20.2	22.5	23.6	22.5	20.1	16.6	12.0	9.1	(27)	
		MCDB	11.9	12.8	17.7	23.8	27.4	30.0	31.8	30.2	25.5	20.4	16.0	11.4	(28)	
	2%	WB	7.4	7.7	10.4	14.4	18.6	21.2	22.0	21.3	18.6	15.1	10.8	7.8	(29)	
		MCDB	9.9	11.0	15.0	21.3	25.2	28.1	29.2	28.5	23.0	17.9	13.4	9.9	(30)	
	5%	WB	5.9	6.5	9.2	13.1	17.2	20.0	20.8	20.4	17.5	14.0	9.8	6.6	(31)	
		MCDB	8.0	9.1	13.0	19.0	22.7	25.9	27.6	26.8	21.1	16.3	12.1	8.4	(32)	
	10%	WB	4.5	5.3	8.0	11.8	15.8	18.7	19.8	19.5	16.5	13.0	8.6	5.3	(33)	
		MCDB	6.2	7.4	11.1	16.6	20.8	23.5	25.2	25.0	19.4	15.2	10.4	7.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.9	7.7	9.2	9.3	9.0	9.2	9.1	7.6	6.9	4.9	4.5	(35)	
		MCDBR	6.6	8.9	12.1	14.2	14.1	13.3	13.4	13.0	11.2	9.5	8.2	6.8	(36)	
	5% WB	MCWBR	4.4	5.5	6.8	7.6	6.8	5.8	5.5	5.0	5.4	5.3	5.1	4.4	(37)	
		MCWBR	6.2	8.2	11.3	13.6	13.4	12.5	12.5	12.1	10.2	8.3	7.4	6.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.295	0.323	0.364	0.399	0.409	0.414	0.419	0.411	0.383	0.350	0.322	0.294	(40)	
		taud	2.401	2.330	2.286	2.253	2.265	2.292	2.278	2.318	2.371	2.442	2.466	2.442	(41)	
	Ebn at Noon		783	831	848	851	854	851	840	830	820	792	744	739	(42)	
		Edh at Noon	67	90	111	126	130	127	127	117	101	81	64	58	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.15	2.07	3.09	4.19	4.83	5.31	5.26	4.67	3.37	2.26	1.22	0.92	(44)	
	RadStd		0.15	0.27	0.37	0.51	0.57	0.50	0.51	0.53	0.45	0.24	0.17	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (99 neighbors)		+0.50	N/A	N/A	+0.75	+0.42	+0.36	N/A	N/A	+75	+34

Nomenclature: See separate page